

MILOŠ PANIĆ

MECHATRONICS ENGINEER

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PROFILE

Mechatronics Engineer with 8+ years of cross-functional experience in machine design, industrial automation, R&D, commissioning and field engineering. Combines hands-on machine development with on-site troubleshooting to deliver reliable, maintainable systems. Experienced in one-off machinery, robotics, machine vision, retrofit and multidisciplinary engineering integration.

WORK EXPERIENCE

Field Service Engineer (Sep 2025 - Present)

Borncut B.V., Urk, Netherlands

- Perform international installation, commissioning, troubleshooting and technical support for industrial machinery.
- Diagnose mechanical, electrical and automation faults using Beckhoff/TwinCAT, servo drives, IPC/HMI, VFDs and sensors.
- Turn recurring field issues into improvements in reliability, maintainability and serviceability, supported by procedures and technical documentation.

R&D Mechatronic Engineer (Nov 2023 - Nov 2024)

Insist ElektriK, Novi Sad, Serbia

- Designed custom industrial machines from concept and CAD through prototyping, assembly, testing and commissioning.
- Developed a one-off (1-of-1) automated BMW assembly machine for seven product variants using rapid prototyping and iterative validation.
- Worked with ABB, FANUC and Mitsubishi robotics, ROS2, SensoPart vision, CNC machining and 3D printing.

Project Engineer - Mechatronics & Industrial Systems (Mar 2020 - May 2023)

Strojoremont, Vukovar, Croatia

- Coordinated multidisciplinary teams and external contractors across industrial engineering, retrofit and maintenance projects.
- Upgraded legacy rubber extrusion and mixing machinery using PLC, SCADA and VFD systems and modern automation standards.
- Supported multidisciplinary modernization and R&D projects spanning food-processing, HVAC and biomass equipment.

EDUCATION

Bachelor's Degree in Mechatronics Engineering (240 ECTS)

Faculty of Technical Sciences, University of Novi Sad, Serbia

Specialization: Automotive Mechatronics

Bachelor thesis: Design, Operation and Control of the ZF 722.6 5G-Tronic Automatic Transmission.

KEY SKILLS

- Machine Design: Mechanical design, custom machinery, fixtures, design for manufacturing, engineering drawings.
- Automation & Controls: Beckhoff TwinCAT, Siemens S7, PLC, VFD, HMI, sensors, commissioning.
- Robotics & Vision: ABB, FANUC, Mitsubishi, ROS2, SensoPart, Balluff.
- CAD & Prototyping: SolidWorks, Inventor, Siemens NX, AutoCAD, CNC, 3D printing, reverse engineering.
- Engineering Integration: Root cause analysis, reliability, maintainability, troubleshooting, technical documentation.
- Project Engineering: Retrofit, supplier/contractor coordination, customer training, continuous improvement.

Technical School Teacher - Mechanical Engineering & Automation (Oct 2019 - Jul 2023)

Technical School Nikola Tesla, Vukovar, Croatia

- Taught mechanical engineering, automation and automotive mechatronics with focus on practical industrial applications.
- Delivered hands-on CAD, machining, CNC and robotics instruction, strengthening technical training and communication skills.

Field Service Engineer - Automation & Access Control Systems (Sep 2018 - Dec 2019)

CAM Engineering / Access Control Systems, Novi Sad, Serbia

- Installed, commissioned and maintained access-control, parking and safety systems including Siemens S7 PLC-based automation.
- Diagnosed mechanical, electrical and control-system faults and performed practical fabrication, machining and repair work on site.

Mechatronics & Robotics Laboratory Technician (Sep 2017 - Dec 2019)

Faculty of Technical Sciences, Novi Sad, Serbia

- Supported laboratory work in robotics, industrial automation and manufacturing using ABB, FANUC and Mitsubishi robots, CNC equipment and 3D printers.
- Supported Eurobot and student engineering projects with mechanical development, prototyping, manufacturing and system integration.

PORTFOLIO milospanic.com

CERTIFICATIONS

- Hybrid Bonding Systems | Hirata Engineering Germany / Hirata Corp Japan (2023)
- Siemens Milling Machines | Vukovar (2020)

KEY PROJECTS

- BMW One-Off Automated Assembly Machine: Designed a purpose-built 1-of-1 automated machine for seven product variants, covering concept, mechanical design, rapid prototyping, integration and validation.
- Grundfos Multi-Variant QC Test Station: Designed a final quality-control station for approximately 20 pump variants using SensoPart machine vision and precision sensors.
- Industrial Retrofit & Modernisation: Developed retrofit concepts for legacy industrial equipment, integrating modern controls, improved maintainability and updated engineering standards.
- Biomass Processing Equipment: Participated in development and manufacturing of biomass extruder presses and dehumidification systems.

LANGUAGES

- Serbian / Croatian - Native
- English - C1
- German - A2